Optics Letters

Real-time measurement of temperature distribution inside a gain medium of a diode-pumped Er^{3+}/Yb^{3+} 1.55 μm laser

Chenchen Xu,[1,2] Yidong Huang,[1] Yanfu Lin,[1] Jianhua Huang,[1] Xingxong Gong,[1] Zundu Luo,[1] and Yujin Chen[1,*]

[1]Key Laboratory of Optoelectronic Materials Chemistry and Physics, Fujian Institute of Research on the Structure of Matter, Chinese Academy of Sciences, Fuzhou, Fujian 350002, China
[2]Fujian Normal University, Fuzhou, Fujian 350007, China
*Corresponding author: cyj@fjirsm.ac.cn

Received 12 July 2017; accepted 28 July 2017; posted 3 August 2017 (Doc. ID 302215); published 24 August 2017

A scheme to measure temperature distribution inside a gain medium of a diode-pumped solid-state laser is proposed based on the fluorescence intensity ratio technique. By recording the temperature-dependent upconversion fluorescence related to the $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transitions of Er^{3+}, the real-time temperature distribution inside an Er^{3+}:Yb^{3+}:$LuAl_3(BO_3)_4$ crystal end-pumped by a 976 nm diode laser can be measured while a 1.55 μm laser is operating. Influences of pump power, output power, and output mirror transmissivity on the temperature distribution inside the gain medium are investigated. This scheme can conveniently and accurately measure the temperature distribution along the laser path in a rare-earth-doped gain medium in real time. © 2017 Optical Society of America

OCIS codes: (140.3480) Lasers, diode-pumped; (140.3500) Lasers, erbium; (140.3580) Lasers, solid-state; (140.6810) Thermal effects.

https://doi.org/10.1364/OL.42.003383

Due to the quantum defects caused by energy differences between pump and laser photons, as well as multiphonon nonradiative transitions between some related multiplets of doped active ions, pump-induced heat is inevitably generated in gain media of solid-state lasers [1,2]. The pump-induced thermal effects limit the output power, operation efficiency, and beam quality and even cause the fracture of gain media [1,3]. Temperature distribution can directly reflect the thermal effects in gain media. Therefore, when a laser is operating, the accurate and real-time measurement of temperature distribution inside the gain medium is meaningful. It is conducive to designing laser devices for reducing the thermal effects and then enhancing the laser performance.

Surface temperatures of gain media can be measured directly by infrared cameras [3,4]. However, this method cannot measure the internal temperatures of gain media and the measured values are easily disturbed by the infrared signals reflected from other surfaces. Temperature distributions inside gain media can be simulated theoretically based on the finite-element-analysis method [5,6]. However, because some hypotheses are adopted in the establishment of the theoretical model and some uncertain thermo-optic coefficients are used in the simulating process, the result may have a deviation from the actual situation.

Utilizing the temperature-dependent feature of fluorescence spectra originating from two thermally coupled levels (TCLs) of some rare earth (RE) ions, the fluorescence intensity ratio (FIR) technique can realize a temperature measurement at the luminous point, where FIR represents the ratio of the integrated intensities I of the two fluorescence bands related to the TCLs. This technique can be used in harsh environments, such as high-voltage power stations and oil refineries [7,8], with its unique advantages of noncontact, high accuracy and sensitivity, and excellent anti-interference. For a diode-pumped solid-state laser with high pump power density, upconversion (UC) visible fluorescence is often observed in a gain medium doped with RE ions, such as Nd^{3+}, Yb^{3+}, Er^{3+}, Tm^{3+}, and Ho^{3+} [9–11]. UC fluorescence is easily detected because it can be seen by naked eyes and is generally away from the pump and laser wavelengths. Then, by measuring the temperature-dependent UC fluorescence intensities originating from two TCLs of RE ions along the laser path in the gain medium, the temperature distribution along the path can be obtained when a laser is operating.

The $^2H_{11/2}$ and $^4S_{3/2}$ levels of Er^{3+} with energy separations of about 780 cm^{-1} are well-known TCLs [7,8]. UC green fluorescence bands originating from the $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ transitions of Er^{3+} can be observed under excitation at 980 nm for an Er^{3+}/Yb^{3+}-codoped material via the energy transfer from Yb^{3+} to Er^{3+} [8,10]. Presently, temperature measurements of Er^{3+}/Yb^{3+}-codoped optical fibers and phosphors as optical temperature sensors and heat-sensitive materials have been widely investigated based on the FIR technique [8,12]. In this work, we report on the first successful measurement of real-time temperature distribution along the

laser path inside an Er^{3+}:Yb^{3+}:$LuAl_3(BO_3)_4$ (Er:Yb:LuAB) crystal, which has been demonstrated as an efficient gain medium of a diode-pumped Er^{3+}/Yb^{3+} 1.55 μm laser [2,13], while the laser is operating.

A *c*-cut, 0.7 mm thick Er(1.1 at. %):Yb(24.1 at. %):LuAB crystal with cross section of 5 mm × 5 mm was polished and used in the spectral and laser experiments. To obtain a relationship between the intensity ratio of UC fluorescence bands originating from the TCLs and the crystal temperature, the crystal was placed in a heating stove with a temperature control range from 300 to 540 K and an accuracy of ±0.1 K. Unpolarized UC fluorescence of the whole crystal was collected and detected from the surface with a cross section of 0.7 mm × 5 mm when a 980 nm laser diode (LD) (MDL-III-980, CNI) was used as the exciting source. In order to ensure a low thermal loading generated by the LD excitation and the uniformity of temperature in the whole sample, the pump power was limited to 0.5 W and the spot diameter of the pump beam was about 5 mm. Then, the UC fluorescence spectra from 510 to 570 nm of the Er:Yb:LuAB crystal were recorded by a spectrometer (FLS920, Edinburgh) in a temperature interval of 30 K and are shown in Fig. 1(a). There are two discrete bands around 520 and 540 nm, corresponding to the transitions of $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$, respectively. For the sake of brevity, the transitions of $^2H_{11/2} \rightarrow {}^4I_{15/2}$ and $^4S_{3/2} \rightarrow {}^4I_{15/2}$ are represented by 520 and 540 hereinafter, respectively.

Because the populations at the $^2H_{11/2}$ and $^4S_{3/2}$ TCLs obey the Boltzmann thermal equilibrium, the relationship between the FIR, which is the ratio of I_{520} and I_{540} in this work, and the crystal temperature T can be expressed as [7,12]

$$\mathrm{FIR} = \frac{I_{520}}{I_{540}} = \frac{\sigma_{520}\omega_{520}g_{520}}{\sigma_{540}\omega_{540}g_{540}} \exp\left(-\frac{\Delta E}{kT}\right), \quad (1)$$

where σ is the emission cross section, ω is the angular frequency of the emission light, g is the degeneracy, k is the Boltzmann constant and ΔE is the energy gap between the $^2H_{11/2}$ and $^4S_{3/2}$ levels. Figure 1(b) shows the integrated intensities of fluorescence bands around 520 and 540 nm for different crystal temperatures. Then, values of the FIR for different crystal temperatures can be calculated, as shown in Fig. 1(c). Based on Eq. (1), the relationship between the FIR and the crystal temperature was fitted to be

$$\mathrm{FIR} = 8.67 \exp\left(-\frac{1083.6}{T}\right). \quad (2)$$

The fitted energy gap ΔE of the Er:Yb:LuAB crystal is 753 cm^{-1} and close to the 780 cm^{-1} of the Er^{3+}:YAB crystal, which was fitted from the low-temperature spectra [14]. The relative temperature sensitivity S of FIR for the Er:Yb:LuAB crystal is expressed by [7,12]

$$S = \frac{1}{\mathrm{FIR}}\frac{\partial \mathrm{FIR}}{\partial T} = \frac{\Delta E}{kT^2}. \quad (3)$$

The maximum S value for the crystal is about 1.2% K^{-1} at 300 K and larger than those of many Er^{3+}-doped thermometric materials (generally lower than 1.1% K^{-1}) [7,8], which shows that the Er:Yb:LuAB crystal has an excellent signal discriminability for temperature detection.

The laser experimental setup is shown in Fig. 2. A continuous-wave (cw) 976 nm fiber-coupled diode laser with a 100 μm diameter core was used as the pumping source, which excites the Yb^{3+} and Er^{3+} ions to the same levels as the 980 nm LD used above. The pumping beam was focused to a spot with a waist diameter of 100 μm in the Er:Yb:LuAB crystal by using a telescopic lens system (TLS). The uncoated Er:Yb:LuAB crystal and a 1.2 mm thick sapphire crystal were placed in a copper holder and closely contacted by two copper pieces with screws at front and rear faces. The copper holder was cooled by water at 293 K. The sapphire crystal was used as a heat sink to reduce the thermal effects of the gain medium and then improve the output laser performances [2,15]. An input mirror (IM) with 90% transmissivity at 976 nm and 99.8% reflectivity at 1.55 μm was directly deposited onto the front surface of the sapphire crystal. Two output mirrors (OMs) with identical curvature radii of 100 mm but different transmissivities of 1.7% and 5.0% at 1.55 μm were used. The cavity length was close to 100 mm. By using a convex lens with a focal length of 27 mm, an image of the UC fluorescence signal emitting from a 0.7 mm × 5 mm surface of the Er:Yb:LuAB crystal was obtained with a magnification of 1.4. A high-resolution optical fiber spectrometer with a fiber diameter of 100 μm (QE65000, Ocean Optics) was used to scan the image and record the fluorescence signal point by point when a 1.55 μm laser was operating. Scattering signals originating from the pump and fundamental lasers were obstructed by a filter.

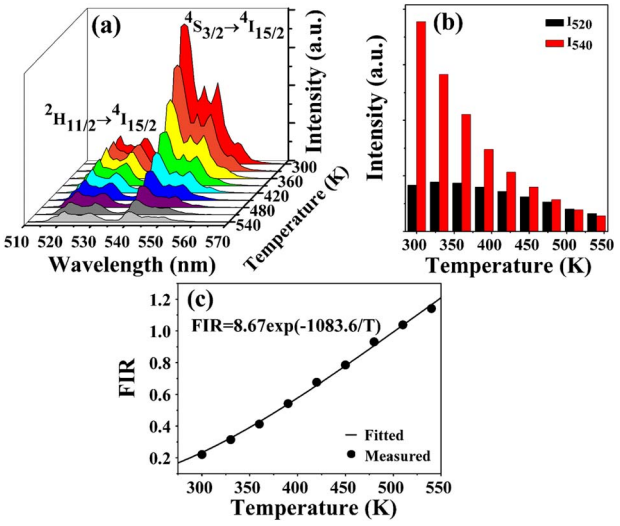


Fig. 1. (a) Temperature-dependent UC fluorescence spectra of the Er:Yb:LuAB crystal recorded from 300 to 540 K when a 980 nm LD was used as the exciting source. (b) Integrated intensities I of fluorescence bands around 520 and 540 nm at different crystal temperatures. (c) FIR between 520 and 540 nm fluorescence bands as a function of crystal temperature.

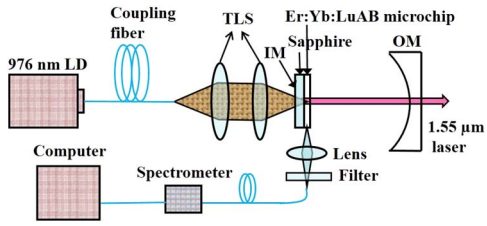


Fig. 2. Experimental setup of the 976 nm diode-pumped Er:Yb:LuAB 1.55 μm laser, and the UC fluorescence measurement.

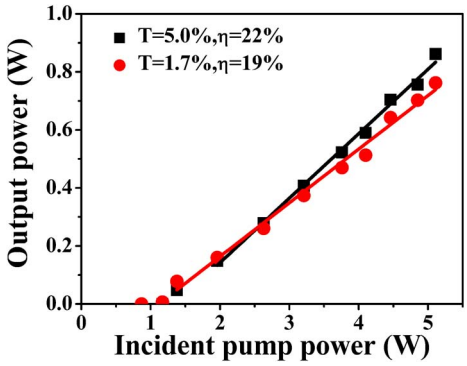


Fig. 3. Output power realized in the Er:Yb:LuAB crystal as a function of an incident pump power at 976 nm for different OM transmissivities T.

Figure 3 shows the cw output power realized in the Er:Yb:LuAB crystal versus incident pump power for different OM transmissivities. For the 5.0% transmissivity, a laser with a maximum output power of 0.86 W and a slope efficiency of 22% was obtained at an incident pump power of 5.11 W. The incident pump threshold was 1.17 W. For the 1.7% transmissivity, a laser with a maximum output power of 0.76 W and a slope efficiency of 19% was obtained at an incident pump power of 5.11 W. The incident pump threshold was 0.87 W. The beam quality factors M_x^2 and M_y^2 were measured to be about 5.71 and 5.33 at a pump power of 5.11 W, respectively, and they changed with the pump power.

When the pump power was 5.11 W and OM transmissivity was 5.0%, UC fluorescence signals emitting from different positions along the laser path with an interval of 0.07 mm were detected through moving the probe of the optical fiber spectrometer along the crystal image after the convex lens. The result is shown in Fig. 4(a). Based on Eq. (2), the corresponding temperature at each position can be calculated with a maximum error of ±12 K and is shown in Fig. 4(b). At a pump power of 5.11 W, the highest temperature in the crystal is 710 K at the position of 0.35 mm from the input surface. The temperature at the input surface of the Er:Yb:LuAB crystal is 450 K and about 100 K lower than that at the output surface. Under identical experimental conditions, temperature distributions for pump powers of 3.74 and 2.63 W were also measured and are also shown in Fig. 4(b). Compared with the result obtained at a pump power of 5.11 W, the trend of temperature distribution inside the crystal is similar but the highest temperatures decrease 150 and 220 K, respectively.

The temperature distribution measured in this work is similar to that inside the Er:Yb:YAB crystal calculated by the finite-element-analysis method [15], in which the highest temperature in the crystal with sapphire cooling was 800 K when the pump power was 5 W. According to [15], the temperature distribution in the crystal without sapphire cooling is similar to the distribution of the pump power density [see Fig. 4(c)]. For diode-pumped Nd:YAG and Yb:YAG lasers, the reported highest temperatures in these crystals were 393 and 335 K with pump power densities of 191 and 318 W/mm^2 [4,6], respectively. They are much lower than that (491 K) measured in this work when the pump power density is 335 W/mm^2, which corresponds to an incident pump

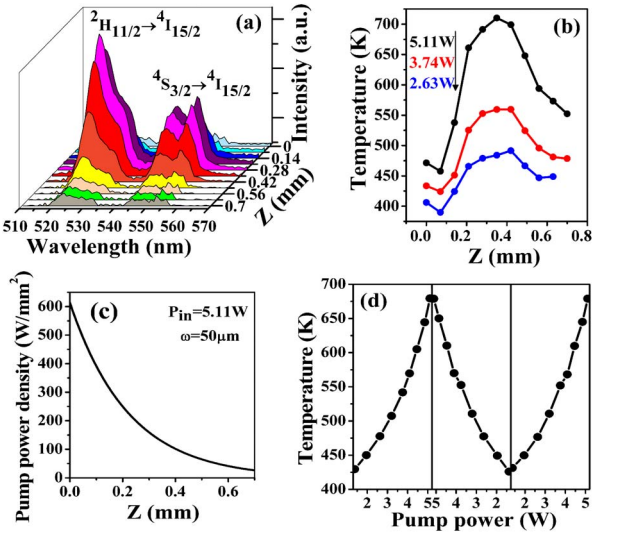


Fig. 4. (a) UC fluorescence spectra at different positions Z along the laser path from the input surface of the Er:Yb:LuAB crystal for a pump power of 5.11 W and an OM transmissivity of 5.0%. (b) Temperature distribution along the laser path inside the Er:Yb:LuAB crystal for different pump powers. (c) Estimated distribution of the pump power density in the Er:Yb:LuAB crystal for a pump power of 5.11 W and a waist diameter of 100 μm. (d) Cyclic measurements of the temperature at the position of 0.28 mm from the input surface of the Er:Yb:LuAB crystal when the pump power was changed back and forth.

power of 2.63 W. Compared with the Nd^{3+} and Yb^{3+} lasers, more absorbed pump power is converted to heat for the Er:Yb:LuAB laser because it has a larger quantum defect (37% with respect to 24% of an 807 nm diode-pumped Nd^{3+} 1.06 μm laser and 6% of a 976 nm diode pumped Yb^{3+} 1.04 μm laser, respectively) and a lower fluorescence quantum efficiency of the upper laser level $^4I_{13/2}$ (7% with respect to about 90% of both Nd^{3+}:YAG and Yb^{3+}:YAG crystals).

Cyclic measurements of the temperature at the position of 0.28 mm from the input face of the Er:Yb:LuAB crystal, where the strongest UC fluorescence signal was observed, were performed by changing the pump power. The result indicates that the temperature measurement in this work has a good repeatability, as shown in Fig. 4(d).

At a certain pump power, the output power of the laser can be gradually changed by adjusting the cavity alignment. Temperature variation at the position of 0.28 mm was measured when the pump power was fixed at 5.11 W, but the output power was changed. As shown in Fig. 5, when the output power was reduced from 0.86 to 0.17 W, the temperature at the measured position increased from 694 to 870 K. The fractional thermal load η_h can be expressed as [1]

$$\eta_h = 1 - \eta_p\left[(1-\eta_l)\eta_r\frac{\lambda_p}{\lambda_f} + \eta_l\frac{\lambda_p}{\lambda_l}\right], \tag{4}$$

where η_p is the pump quantum efficiency, η_l is the laser extraction efficiency defined as the fraction of excited ions extracted by stimulated emission, η_r is the fluorescence quantum efficiency of the upper laser level, λ_p is the pump wavelength,

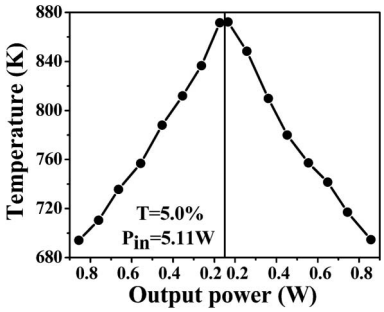


Fig. 5. Cyclic measurements of the temperature at the position of 0.28 mm from the input surface of the Er:Yb:LuAB crystal when the output power was decreased and increased at a fixed pump power of 5.11 W and an OM transmissivity of 5.0%.

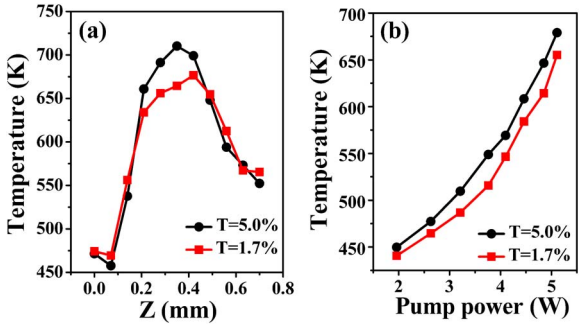


Fig. 6. (a) Temperature distributions inside the Er:Yb:LuAB crystal for different OM transmissivities when the pump power was 5.11 W. (b) Temperatures at the crystal position of 0.28 mm for different pump powers and OM transmissivities.

λ_f is the average fluorescence wavelength, and λ_l is the laser wavelength. Because η_r of the Er:Yb:LuAB crystal is lower than 10% [2], η_h in the crystal increases with the decrement of η_l, which is proportional to the reduction of the output power at a fixed pump power. Therefore, with the reduction of the output power, the temperature inside the crystal increases, as shown in Fig. 5. It also means that an efficient laser operation would be beneficial in reducing the fractional thermal load in the crystal.

The temperature distribution at a pump power of 5.11 W and the temperature at the position of 0.28 mm for different pump powers were also measured at OM transmissivity of 1.7%. The results obtained at OM transmissivities of 1.7% and 5.0% are compared in Figs. 6(a) and 6(b), which exhibit similar variation trends with maximum measurement errors of ±12 K. With the increment of the pump threshold, the heat generated in the crystal is increased. Therefore, the highest temperature inside the crystal at an OM transmissivity of 5.0% is about 40 K higher than that at an OM transmissivity of 1.7%, as shown in Fig. 6(a), while the pump thresholds are 1.17 and 0.87 W for the former and later, respectively.

In summary, a measurement of temperature distribution along laser path inside an Er:Yb:LuAB crystal has been successfully realized when a 1.55 μm laser is operating. The highest temperature inside the crystal is 710 K when the pump power is 5.11 W and the output power is 0.86 W. Compared with the existing thermometries, this method can realize the real-time temperature measurement inside a gain medium with advantages of high accuracy and excellent anti-interference. This method may be used to measure the temperature distributions along any direction inside RE-doped gain media with UC fluorescence emitting from TCLs during laser operation.

Funding. National Key R&D Program of China; Ministry of Science and Technology of the People's Republic of China (MOST) (2016YFB0701002); Strategic Priority Research Program, Chinese Academy of Sciences (CAS) (XDB20000000).

REFERENCES

1. T. Y. Fan, IEEE J. Quantum Electron. **29**, 1457 (1993).
2. Y. Chen, Y. Lin, J. Huang, X. Gong, Z. Luo, and Y. Huang, Opt. Express **23**, 12401 (2015).
3. S. Chénais, F. Druon, S. Forget, F. Balembois, and P. Georges, Prog. Quantum Electron. **30**, 89 (2006).
4. J. Didierjean, E. Herault, F. Balembois, and P. Georges, Opt. Express **16**, 8995 (2008).
5. E. Marmois, V. Cardinali, G. L. Touzé, and B. L. Garrec, Proc. SPIE **7721**, 77210O (2010).
6. P. A. Loiko, K. V. Yumashev, N. V. Kuleshov, and A. A. Pavlyuk, Opt. Laser Technol. **44**, 2232 (2012).
7. S. A. Wade, S. F. Collins, and G. W. Baxter, J. Appl. Phys. **94**, 4743 (2003).
8. X. Wang, Q. Liu, Y. Bu, C.-S. Liu, T. Liu, and X. Yan, RSC Adv. **5**, 86219 (2015).
9. V. Ostroumov, T. Jensen, J.-P. Meyn, G. Huber, and M. A. Noginov, J. Opt. Soc. Am. B **15**, 1052 (1998).
10. F. Song, L. Han, H. Tan, J. Su, J. Yang, J. Tian, G. Zhang, Z. Cheng, and H. Chen, Opt. Commun. **259**, 179 (2006).
11. G. Hansson, A. Callenas, and C. Nelsson, *Advanced Solid-State Lasers* (Optical Society of America, 1993), p. 446.
12. L. Li, L. Zheng, W. Xu, Z. Liang, Y. Zhou, Z. Zhang, and W. Cao, Opt. Lett. **41**, 1458 (2016).
13. Y. J. Chen, Y. F. Lin, J. H. Huang, X. H. Gong, Z. D. Luo, and Y. D. Huang, Opt. Express **18**, 13700 (2010).
14. I. Földvári, E. Beregi, A. Munoz, F. R. Sosa, and V. Horváth, Opt. Mater. **19**, 241 (2002).
15. Y. J. Li, J. X. Feng, P. Li, K. S. Zhang, Y. J. Chen, Y. F. Lin, and Y. D. Huang, Opt. Express **21**, 6082 (2013).